

SHANGQI GAO, PH.D.

Early Cancer Institute, University of Cambridge, CB2 0XZ, Cambridge, UK

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ACADEMIC EXPERIENCE

Research Associate

University of Cambridge

Mar 2024 – Present

Cambridge, UK

- AI for Cancer Imaging integrating machine learning and multi-omics data
- Affiliated with the Early Cancer Institute, Department of Oncology
- Supervisor: Dr. Mireia Crispin-Ortuzar (mc973@cam.ac.uk)

Postdoctoral Research Assistant

University of Oxford

Apr 2023 – Feb 2024

Oxford, UK

- Urological cancer pathology AI and uncertainty-aware deep learning
- Affiliated with the Nuffield Department of Surgical Science
- Supervisor: Prof. Clare Verrill (Clare.Verrill@ouh.nhs.uk)
- Co-Supervisor: Prof. Jens Rittscher (jens.rittscher@eng.ox.ac.uk)

EDUCATION

Ph.D. in Statistics

School of Data Science, Fudan University

Sep 2018 – Jul 2022

Shanghai, China

- Thesis: *Deep Image Decomposition and Reconstruction*
- Advisor: Prof. Xiahai Zhuang (zxh@fudan.edu.cn)

M.Sc. in Applied Mathematics

School of Mathematics and Statistics, Wuhan University

Sep 2015 – Jul 2018

Wuhan, China

- Thesis: *Regularization-Based Approaches for Tensor Completion*
- Advisor: Prof. Qibin Fan

B.Sc. in Applied Mathematics

Department of Mathematics, Northwestern Polytechnical University

Sep 2011 – Jul 2015

Xi'an China

- Thesis: *Fourier Analysis and Its Applications in Solving Differential and Integral Equations*
- Advisor: Prof. Pengcheng Niu

AWARDS & HONORS

Best Paper Award Candidate, CVPR

2026

Best Paper Award, MICCAI AMAI

2025

Shanghai Natural Science Award (2/5), Shanghai Government

2023

Elsevier–MedIA 1st Prize & Medical Image Analysis Best Paper Award, MICCAI

2023

Excellent Graduate Award, Shanghai Higher Education Department

2022

Best Paper Finalist, MICCAI

2022

National Graduate Scholarship (Ph.D.), Fudan University

2021

National Graduate Scholarship (M.Sc.), Wuhan University

2017

Meritorious Winner of MCM, SIAM

2014

Honorable Mention of MCM, CSIAM	2013
National Endeavor Scholarship, Northwestern Polytechnical University	2012

PROJECTS PARTICIPATED

Postdoctoral period (led/participated projects):	
Multi-omics Data Integration for Ovarian Carcinoma (Principal Investigator) 10,000 GPU hours (£5,500), DAWN Pioneer Project, Cambridge, UK	2025 – 2026
Computational Pathology and Multi-omics Data Integration for Ovarian Carcinoma (Co-led) 40,000 GPU hours (£22,000), EPSRC Tier-2 HPC, Cambridge, UK	2024 – 2025
Longitudinal Image and Volumetric Analysis, Visualisation and Measurements (Research Associate) GEHC-Cambridge, Cambridge, UK	2024 – 2026
Urological Cancer Pathology AI - Beyond Prostate (Postdoctoral Research Assistant) Medical Research Council, Oxford, UK	2023 – 2024
Doctoral period (participated national/international projects led by Prof. Xiahai Zhuang):	
Methodologies and Applications of Combined Segmentation of Multimodality Cardiac Images Based on Multivariate Mixture Model National Natural Science Foundation of China (NSFC), Shanghai, China	2020 – 2023
Self-supervised Deep Learning for Multimodal Whole Heart Segmentation of Four-dimensional Cine Images NSFC-STINT (China–Sweden Joint Research Grant), Shanghai, China	2021 – 2023
Cardiac Image Segmentation via Deep Learning with Regularization using Prior Sub Networks NSFC-NRF (China–Korea Joint Research Grant), Shanghai, China	2020 – 2022

TEACHING & SUPERVISION

Supervisor , <i>Why Should I Trust You: Explainable AI in Cancer Imaging</i> , MPhil Data Intensive Science Project MPhil student, Department of Physics, University of Cambridge	Lent & Easter Terms, 2025
Guest Lecturer , <i>Image Processing and Analysis</i> , Summer Programme Homerton College, University of Cambridge	Aug 2024
Supervisor , <i>Training-free OOD Medical Tumor Segmentation via Anatomical Reasoning and Statistical Rejection</i> PhD candidate, School of Computer Science, Carnegie Mellon University	2025 – 2026
Supervisor , <i>Towards Generalizable Retinal Vessel Segmentation with Deformable Graph Priors</i> PhD candidate, School of Computer Science, Zhejiang University	2024 – 2025
Co-supervisor , <i>Multi-modal MRI Translation via Evidential Regression and Distribution Calibration</i> PhD candidate, TSTBI, Fudan University	2024 – 2025
Co-supervisor , <i>InDeed: Interpretable image deep decomposition with guaranteed generalizability</i> PhD candidate, School of Data Science, Fudan University	2023 – 2024
Guest Lecturer , <i>DATA630015 Medical Image Analysis</i> , Postgraduate Course School of Data Science, Fudan University, China	2020 – 2021
Teaching Assistant , <i>DATA130012.01 Data Visualization</i> , Postgraduate Course School of Data Science, Fudan University, China	2019 – 2020
Teaching Assistant , <i>Sparsity in Statistics</i> , Postgraduate Course School of Data Science, Fudan University, China	2018 – 2019

PRESENTATIONS & TALKS

Invited talk: <i>Explainable Integration of Radiology and Pathology for Pan-Cancer Analysis</i> Early Cancer Institute, University of Cambridge	July 2026
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Invited talk: <i>Probabilistic Modelling and Bayesian Deep Learning in Multimodal Data Integration</i> Hawkes Institute, University College London	Apr 2026
Oral: <i>Towards Generalizable Retinal Vessel Segmentation with Deformable Graph Priors</i> NeurIPS@Cam Early Career Researcher Conference, University of Cambridge	Dec 2025
Poster: <i>Can Foundation Segmentation Models Generalize to Breast Cancer?</i> ESMO AI & Digital Oncology, Berlin, Germany	Nov 2025
Invited talk: <i>Explainable Integration of Kidney Cancer Radiology and Pathology</i> Jeffrey Cheah Biomedical Centre, University of Cambridge	Oct 2025
Poster: <i>Probabilistic Integration of Renal Cancer Radiology and Pathology Using Graph Neural Networks</i> MICCAI, Daejeon, Korea	Sep 2025
Oral: <i>Evaluating Foundation Models with Pathological Concept Learning for Kidney Cancer</i> MICCAI AMAI, Daejeon, Korea	Sep 2025
Oral: <i>Bayesian Modeling for Medical Image Segmentation with Interpretable Generalizability</i> Robust Cancer Early Detection Systems under Distribution Shifts and Uncertainty Workshop, University of Cambridge	Jun 2024
Invited talk: <i>Bayesian Modeling for Medical Image Segmentation with Interpretable Generalizability</i> The 1st Youth Academic Forum on Intelligent Imaging, Chinese Society of Stereology	Apr 2023
Poster: <i>Joint Modelling of Image and Label Statistics for Enhancing Model Generalizability of Medical Image Segmentation</i> MICCAI, Singapore	Sep 2022
Invited talk: <i>Bayesian Deep Learning: Its Applications in Natural Image Super-Resolution and Medical Image Segmentation</i> The 1075th Academic Seminar on Biomedical and Health Engineering, Shenzhen	Jun 2022
Seminar: <i>Rank-One Network: An Efficient Framework for Image Restoration</i> Applied Mathematics Ph.D. Seminar, School of Mathematics, Fudan University	Jun 2021

SELECTED PUBLICATIONS

1. S Shen, K Liu, J Xie, S Gao, C Shen, G Liu, M Crispin-Ortuzar, **S Gao**. *R2Seg: Training-free OOD Medical Tumor Segmentation via Anatomical Reasoning and Statistical Rejection*. CVPR 2026 (Oral, Best Paper Award Candidate).
2. K Liu, S Gao, Y Fu, **S Gao**. *Towards Generalizable Retinal Vessel Segmentation with Deformable Graph Priors*. NeurIPS 2025.
3. **S Gao**, H Zhou, Y Gao & X Zhuang. *BayeSeg: Bayesian Modeling for Medical Image Segmentation with Interpretable Generalizability*. Medical Image Analysis, 2023, 89(7):102889 (Elsevier-MedIA 1st Prize & Medical Image Analysis MICCAI Best Paper Award 2023).
4. **S Gao** & X Zhuang. *Bayesian Image Super-Resolution with Deep Modelling of Image Statistics*. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2023. 45(2): 1405-1423.
5. **S Gao** & X Zhuang. *Rank-One Network: An Effective Framework for Image Restoration*. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2022, 44(6): 3224-3238.

PUBLICATIONS

1. K Liu, S Gao, Y Fu, **S Gao**, C Shen. *Probabilistic Modelling of Multi-rater Medical Image Segmentation for Diversity and Personalization*. preprint, 2025.
2. S Shen, K Liu, J Xie, S Gao, C Shen, G Liu, M Crispin-Ortuzar, **S Gao**. *R2Seg: Training-free OOD Medical Tumor Segmentation via Anatomical Reasoning and Statistical Rejection*. CVPR 2026 (Oral, Best Paper Award Candidate).

3. K Liu, S Gao, Y Fu, **S Gao**. *Towards Generalizable Retinal Vessel Segmentation with Deformable Graph Priors*. NeurIPS 2025.
4. **S Gao**, S Gao, I Machado, M Crispin-Ortuzar. *Probabilistic Integration of Renal Cancer Radiology and Pathology Using Graph Neural Networks*. MICCAI 2025 (Early Accepted, Top 9%).
5. Y Liu, R Zhen, **S Gao**, X Luo, X Gao, Q Chen, and X Zhuang. *BayeSMM: Robust Deep Combined Computing Tackling Heavy-tailed Distribution in Medical Images*. MICCAI 2025 (Oral).
6. J Liu, **S Gao**, Y Li, L Liu, X Gao, Z Xing, J Ning, Y Su, X Zhang, J He, N Xu, and X Zhuang. *Multi-modal MRI Translation via Evidential Regression and Distribution Calibration*. MICCAI 2025 (Early Accepted, Top 9%).
7. Y Gao, H Zhou, Z Gao, B Wang, **S Gao**, S Wang, X Zhuang. *Learning Concept-Driven Logical Rules for Interpretable and Generalizable Medical Classification*. MICCAI 2025 (Young Scientist Award).
8. **S Gao**, S Wang, Y Gao, B Wang, X Zhuang, A Warren, G Stewart, J Jones, and M Crispin-Ortuzar. *Evaluating Foundation Models with Pathological Concept Learning for Kidney Cancer*. MICCAI AMAI 2025 (Oral & Best Paper Award).
9. **S Gao**, N.K. Alham, A Protheroe, K Edwards, J Hamblin, J Rittscher, and C Verrill. *Characterising Borderline Areas in Bladder Tumour Grading with Bayesian Graph Neural Networks*. ISBI 2024 (Oral).
10. **S Gao**, H Zhou, Y Gao & X Zhuang. *BayeSeg: Bayesian Modeling for Medical Image Segmentation with Interpretable Generalizability*. Medical Image Analysis, 2023, 89(7):102889 (Elsevier-MedIA 1st Prize & Medical Image Analysis MICCAI Best Paper Award 2023).
11. **S Gao** & X Zhuang. *Bayesian Image Super-Resolution with Deep Modelling of Image Statistics*. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2023. 45(2): 1405-1423.
12. J Liu, Y Li, N Shi, Y Zhou, **S Gao**, Y Shi, X Zhang and X Zhuang. *Multi-phase Liver-specific DCE-MRI Translation via a Registration-guided GAN*. Simulation and Synthesis in Medical Imaging, 2023: 21-31.
13. K Wan, L Li, D Jia, **S Gao**, W Qian, Y Wu, H Lin, X Mu, X Gao, S Wang, F Wu and X Zhuang. *Multi-target Landmark Detection with Incomplete Images via Reinforcement Learning and Shape Prior Embedding*. Medical Image Analysis, 2023, 89: 102875.
14. **S Gao** & X Zhuang. *Rank-One Network: An Effective Framework for Image Restoration*. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2022, 44(6): 3224-3238.
15. **S Gao**, H Zhou, Y Gao, X Zhuang. *Joint Modelling of Image and Label Statistics for Enhancing Model Generalizability of Medical Image Segmentation*. MICCAI 2022 (Best Paper Finalist).
16. H Zhou, C Huang, **S Gao**, X Zhuang. *VSpSR: Explorable Super-Resolution via Variational Sparse Representation*. CVPR Workshops, 2021: 373-381.
17. **S Gao** & X Zhuang. *Robust Approximations of Low-Rank Minimization for Tensor Completion*. Neurocomputing, 2020, 379: 319-333.
18. **S Gao** & Q Fan. *Robust Schatten-p Norm Based Approach for Tensor Completion*. Journal of Scientific Computing, 82:11, 2020.
19. **S Gao** & Q Fan. *Robust Balancing Scheme-Based Approach for Tensor Completion*. Neurocomputing, 330:328-336, 2019.
20. **S Gao**, X Zhuang. *Multi-Scale Deep Neural Networks for Real Image Super-Resolution*. CVPR Workshops, 2019: 2006-2013.
21. **S Gao** & Q Fan. *A Mixture of Nuclear Norm and Matrix Factorization for Tensor Completion*. Journal of Scientific Computing, 75:43-64, 2018.

ACADEMIC SERVICE

Program Chair , Workshop on Interpretability of Machine Learning in Medical Image Computing MICCAI, Daejeon, Korea	Sep 2025
Co-organiser , Challenge on Comprehensive Analysis & Computing of Real-world Medical Images MICCAI, Daejeon, Korea	Sep 2025
Organiser , Online seminar on Explainable AI for Medical Image Analysis	Jun 2025
Co-organiser , Challenge on Comprehensive Analysis & Computing of Real-world Medical Images MICCAI, Marrakesh, Morocco	Oct 2024
Co-organiser , The 1st Workshop on Multimodal Imaging and Explainable Artificial Intelligence Analysis and International Symposium on Sino-Swedish Medical Image Analysis, Xiamen, China	Jul 2024
Reviewer (selected): Cancer Discovery; Nature Communications; IEEE Transactions on Image Processing; Medical Image Analysis; IEEE Transactions on Medical Imaging; IEEE Transactions on Neural Network and Learning System; Neural Networks; Neurocomputing; NeurIPS; CVPR; ICCV; ECCV; ACM Multimedia; MICCAI.	